

KALACHINSK, Russian FederationWMO: **286960**Lat: **55.0193N**Lon: **74.6125E**Elev: **110**StdP: **100.01**Time Zone: **6.00 (E06)**Period: **94-18**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-34.8	-31.9	-38.1	0.1	-34.8	-35.2	0.1	-31.8	10.2	-9.3	9.2	-9.1	1.4	30	0.634

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.5	30.5	19.0	28.7	18.3	26.9	17.6	20.7	27.5	19.8	26.0	18.8	24.7	4.1	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.5	13.5	23.2	17.5	12.7	22.3	16.6	12.0	21.6	60.4	27.5	57.1	26.0	54.0	24.8	32.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.5	8.4	7.4	-37.0	33.0	4.5	2.2	-40.2	34.6	-42.8	35.8	-45.4	37.0	-48.6	38.6		
(5)			-37.0	22.1	4.5	1.0	-40.3	22.8	-42.9	23.3	-45.4	23.9	-48.7	24.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	1.8	-18.4	-16.1	-7.1	3.9	12.2	17.6	19.0	16.6	10.4	3.3	-6.7	-14.9
	DBStd	14.55	8.07	7.25	5.92	6.18	5.39	4.36	3.72	4.09	4.72	4.85	7.46	8.11
	HDD10.0	3917	879	730	529	200	39	3	0	2	51	213	500	771
	HDD18.3	6197	1138	963	788	434	200	65	38	83	242	467	750	1029
	CDD10.0	908	0	0	0	16	108	230	278	207	63	5	0	0
	CDD18.3	145	0	0	0	0	11	42	59	29	4	0	0	0
	CDH23.3	1549	0	0	0	6	166	448	577	302	50	0	0	0
	CDH26.7	444	0	0	0	1	38	132	173	91	9	0	0	0
Wind	WSAvg	3.2	2.8	2.9	3.3	3.8	3.6	3.0	2.8	2.8	3.0	3.2	3.4	3.2
Precipitation	PrecAvg	361	16	14	15	20	27	51	61	51	27	28	28	22
	PrecMax	548	36	39	37	52	80	114	178	126	65	73	59	45
	PrecMin	242	1	2	1	1	6	16	2	16	8	9	4	8
	PrecStd	74	10	7	10	13	16	23	36	24	18	14	14	9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.5	1.1	6.8	23.5	30.3	32.2	33.0	32.7	27.7	18.7	7.3	1.5
		MCWB	-0.1	0.0	3.2	12.4	16.3	18.6	21.3	19.4	15.5	11.1	5.7	0.4
	2%	DB	-1.9	-1.0	3.9	19.4	27.4	29.9	30.4	29.2	24.5	15.6	5.4	0.3
		MCWB	-2.6	-1.8	1.8	10.5	15.3	18.2	19.7	19.2	14.4	9.4	4.1	-0.4
	5%	DB	-4.0	-2.8	2.2	16.7	25.1	28.0	28.5	26.5	21.0	12.8	3.7	-1.7
		MCWB	-4.7	-3.5	0.8	9.5	14.2	18.1	19.1	17.8	13.5	8.3	2.4	-2.5
	10%	DB	-6.1	-5.0	1.0	13.7	22.3	25.8	26.6	24.4	18.4	10.6	1.7	-3.9
		MCWB	-6.6	-5.7	-0.1	8.0	13.0	17.3	18.7	17.0	12.0	7.1	0.5	-4.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-0.1	0.1	3.5	13.6	18.0	21.3	22.7	21.4	17.6	12.1	6.3	0.7
		MCDB	0.5	0.8	6.1	20.7	27.3	28.4	27.4	28.8	25.8	16.4	7.1	1.1
	2%	WB	-2.6	-1.8	2.0	11.5	16.2	20.0	21.1	19.9	15.5	10.5	4.3	-0.5
		MCDB	-2.0	-1.1	3.4	18.4	25.5	26.4	27.8	26.4	21.9	14.2	5.4	0.3
	5%	WB	-4.6	-3.5	1.0	9.8	14.9	18.9	20.2	18.9	14.0	8.7	2.4	-2.4
		MCDB	-4.0	-2.8	2.1	16.1	23.1	25.5	26.4	24.6	19.6	12.2	3.6	-1.6
	10%	WB	-6.7	-5.8	-0.2	8.2	13.6	17.9	19.3	17.8	12.5	7.2	0.7	-4.7
		MCDB	-6.1	-5.0	1.0	13.4	21.0	24.3	25.1	23.0	17.8	10.4	1.5	-4.0
Mean Daily Temperature Range	5% DB	MDBR	7.3	8.0	8.1	9.4	12.0	11.1	10.5	10.5	9.7	7.4	6.0	7.2
		MCDBR	8.8	6.9	7.0	14.3	15.4	14.3	13.7	14.0	14.0	11.9	6.2	7.6
		MCWBR	8.6	6.7	5.7	7.7	6.8	5.8	5.3	6.1	6.6	7.2	5.5	7.6
	5% WB	MCDBR	8.5	7.1	6.4	13.1	13.9	12.1	11.7	12.0	12.2	10.4	6.1	7.9
		MCWBR	8.3	6.9	5.6	7.5	6.9	5.8	5.5	6.1	6.3	6.8	5.5	7.8
Clear-Sky Solar Irradiance	taub	0.284	0.316	0.376	0.405	0.401	0.410	0.432	0.429	0.368	0.353	0.324	0.288	
	taud	2.043	2.014	1.926	2.127	2.228	2.285	2.216	2.191	2.322	2.231	2.095	2.000	
	Ebn at Noon	609	728	763	811	841	837	807	781	787	689	550	496	
	Edh at Noon	68	102	142	135	129	124	130	126	96	83	66	57	
All-Sky Solar Radiation	RadAvg	0.98	2.07	3.65	4.64	5.57	6.01	5.51	4.44	3.17	1.62	0.78	0.64	
	RadStd	0.07	0.09	0.25	0.36	0.52	0.42	0.37	0.37	0.29	0.20	0.12	0.07	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (13 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page